ACADEMIC PROJECT REPORT

Module: Deep Learning & Artificial Intelligence
Engineering Cycle – 5th year

BCMID

Breast Cancer Multimodal Imaging & Diagnosis

Design and Evaluation of an Intelligent System Supporting Breast Imaging Interpretation

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Applied research project in medical AI · Academic prototype subject to human validation

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Abstract

The **BCMID** project (*Breast Cancer Multimodal Imaging & Diagnosis*) aims to design a prototype supporting the interpretation of breast examinations by integrating heterogeneous imaging data and documentary knowledge. The approach links **full-field digital mammography** (FFDM) and **ultrasound** to an architecture of **seven specialized agents** (data quality, dual-branch ResNet-18 vision with Grad-CAM, FAISS-based clinical retrieval, Qwen2.5-3B-Instruct reasoning, deterministic safety checks and structured reporting), in order to produce a structured, explainable report that is subject to verification.

The system comprises examination preprocessing, two **ResNet-18** branches, **Grad-CAM** maps, a **Clinical RAG** document retrieval system (**BAAI/bge-small-en-v1.5** and **FAISS**), a synthesis agent driven by **Qwen2.5-3B-Instruct** and a deterministic safety layer. Orchestration is handled by **n8n**, in conjunction with **FastAPI** microservices.

On the project corpus (332 patients, 2,001 raw images, 321 PDF reports), the multimodal subset of 323 patients reaches a **ROC-AUC of 0.8198** (“malignant versus others”), versus 0.7917 for mammography alone and 0.6743 for ultrasound alone. The XAI audit reports 87.4% of Grad-CAM energy inside the breast mask; 61/61 software tests pass, 10/10 adversarial scenarios are neutralized and 17/20 reports are compliant on first pass. These results remain exploratory: the annotations do not constitute a histological ground truth, the corpus is limited and no prospective clinical validation is provided. These observations do not establish medical effectiveness.

Keywords: breast imaging, multimodal fusion, explainable deep learning, ResNet-18, Grad-CAM, clinical RAG, FAISS, Qwen2.5, intelligent agents, clinical safety, human oversight.

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General Introduction

Breast cancer is a major public health issue, for which the quality of imaging interpretation plays a decisive role. Breast analysis draws on mammography, ultrasound, clinical data and classification frameworks: these sources are complementary, but their heterogeneity demands a rigorous approach to limit interpretation errors.

Deep learning makes it possible to learn representations useful for classification, and explainable AI methods such as Grad-CAM visualize the regions associated with model decisions. Nevertheless, a high probability is not clinical proof, and an activation map does not confirm the existence of a lesion. Large language models make it easy to produce summaries, but a convincing answer may contain a numerical error or an unsupported claim. It is therefore essential to separate image analysis, knowledge retrieval and report writing, and then to subject the outputs to independent checks.

BCMID combines two ResNet-18 vision branches with Grad-CAM, a *Clinical RAG* document retrieval module (FAISS) and a Qwen2.5-3B-Instruct model, organized into seven agents coordinated by n8n and FastAPI. **The central objective** is to study how a multimodal, explainable and traceable architecture can link visual prediction, documentary justification and report generation while maintaining human supervision. BCMID is a **research and demonstration prototype**, not a validated medical device.

CHAPTER 1

Context and Multi-Agent Architecture

1.1 Clinical Context and Challenges

Full-field digital mammography (FFDM, CC and MLO views) makes it possible to observe microcalcifications and certain architectural distortions; ultrasound characterizes the solid or fluid nature of an abnormality, its contours and its posterior acoustic behavior. Combining the modalities aims to exploit non-identical information. Interpretation is more difficult in the presence of dense tissue, which causes sensitivity to vary [1].

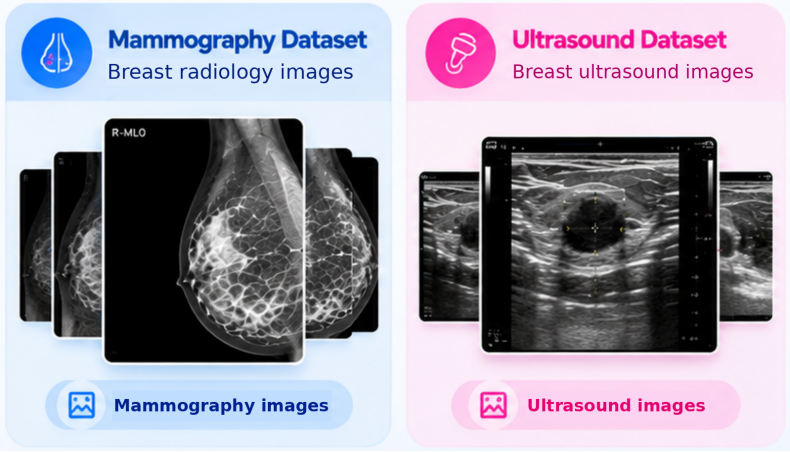


Figure 1.1: Examples of mammography and ultrasound data from the project's image set.

1.2 Problem Statement, Objectives and Scope

How can several medical modalities be analyzed jointly, how can the results be linked to documentary elements, and how can a structured summary be produced whose consistency can be verified before consultation by a specialist? Four challenges follow: **D1 Data heterogeneity**; **D2 Interpretability** (influential regions without presenting them as confirmed lesions); **D3 Documentary reliability** (summaries backed by identifiable excerpts); and **D4 Output safety** (no contradictory or inaccurate content, and none lacking a mention of human review).

Scope. BCMID is an academic interpretation-support prototype, not an approved medical device nor an autonomous diagnostic system. The BI-RADS category derived from a score scale must never be confused with a validated radiological categorization following the ACR Atlas [2]. The field `human_review_required` embodies this restriction.

1.3 Rationale for an Agentic Architecture

BCMID does not delegate all operations to a single generative model. Images are verified before analysis, probabilities come from the vision model, references are retrieved by the RAG and the LLM text is checked before being issued. The split into agents makes ingestion, visual prediction, document retrieval, writing and safety checks independent; sequencing by n8n makes the inputs, outputs and rejection points of each step identifiable. This rationale is architectural: it does not constitute an experimental comparison between several orchestrations.

1.4 Overall Architecture and the Seven Agents

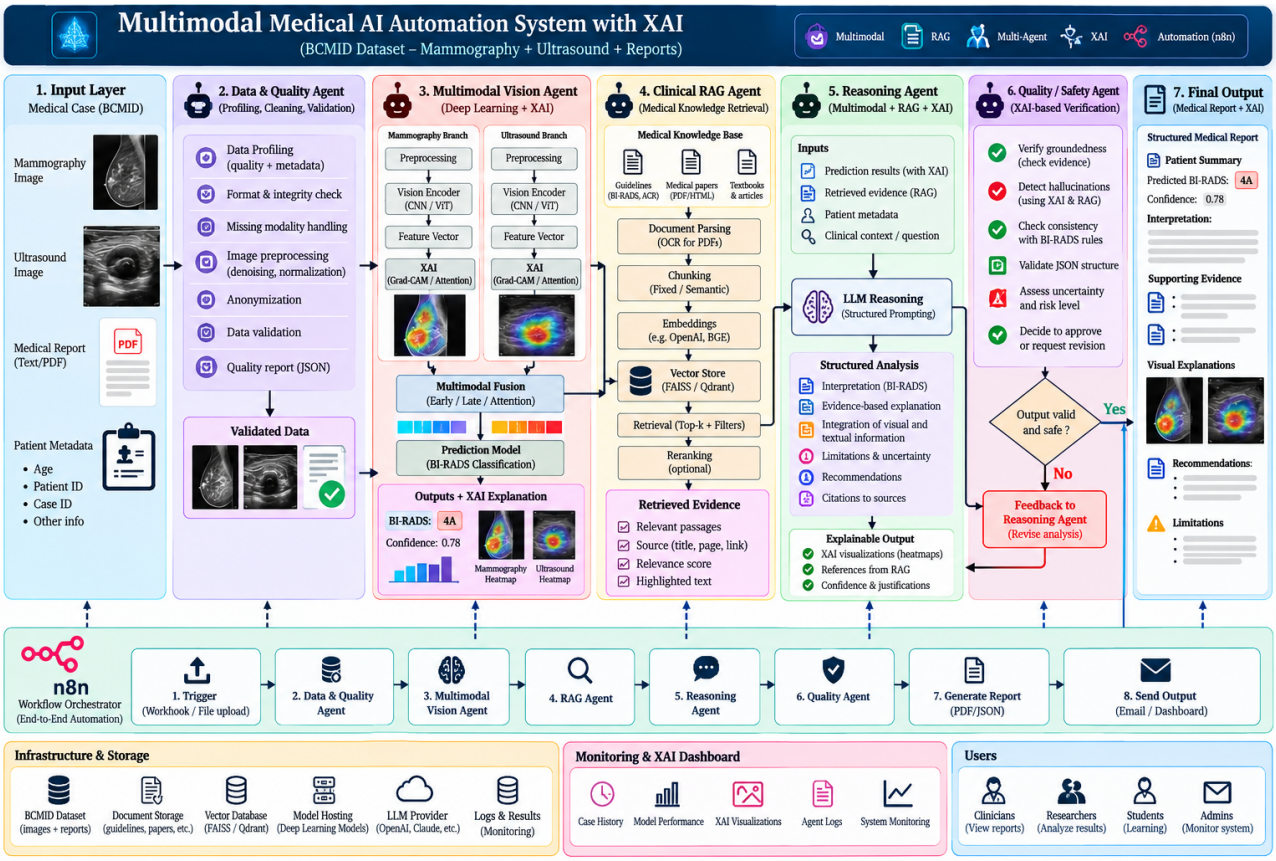


Figure 1.2: BCMID architecture: seven agents, n8n orchestrator, infrastructure, monitoring and users.

Agent	Responsibilities and methods	Technologies	Outputs
01 – Ingestion	Metadata reception and verification, SHA-256 fingerprints, pseudonymization	Python, pandas	Verified record, pseudonymized identifier
02 – Data Quality	Profiling, Otsu, OCR, inpainting, per-patient separation	OpenCV, EasyOCR	Normalized images, quality status
03 – Vision	Two ResNet-18, fusion, scores, Grad-CAM	PyTorch, torchvision	Probabilities, uncertainty, maps
04 – Clinical RAG	Semantic encoding, two FAISS indexes, filtering of excluded records	bge-small-en-v1.5, FAISS IndexFlatIP	Excerpts and references [G1], [E1]
05 – Reasoning	Evidence assembly, constrained generation	Qwen2.5-3B-Instruct	Interpretive JSON proposal
06 – Quality & Safety	Numerical, semantic and documentary validation	Deterministic rules, pytest	Authorization, review or flagging
07 – Reporting	Output in structured formats	ReportLab, FastAPI, Pydantic, n8n	PDF report, JSON, web view

CHAPTER 2

Data and Multimodal Analysis

2.1 Medical Corpus and Annotation Distribution

The **patient level** is the main grouping unit, in order to preserve independence between the training and validation sets.

Set	Volume	Details
Registered patients	332	of which 327 with a supervised label
Raw medical images	2,001	before quality checks
Retained images	1,974	951 mammograms and 1,023 ultrasounds
Medical PDF reports	321	320 kept for indexing after filtering
Patients with both modalities	323	cohort of the ROC-AUC comparison
Total volume	12.11 GB	stated in the source report

The categories are *normal* (15.1%), *benign* (64.5%), *malignant* (19.0%) and five unlabeled BI-RADS 0 cases (1.5%). The predominance of benign cases limits the interpretation of aggregated metrics. The supervised variable is **derived from BI-RADS annotations** and not from independent histological confirmation: the study is an evaluation on these project annotations.

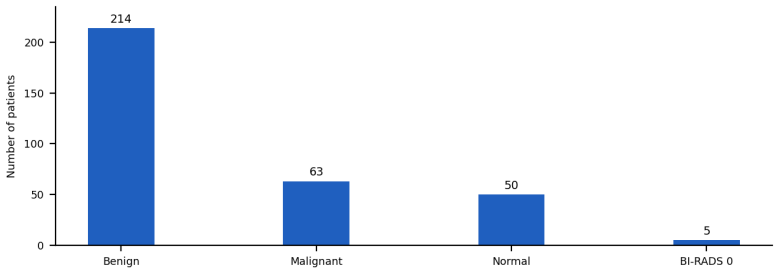


Figure 2.1: Distribution of the 332 patients by annotation category.

2.2 Preprocessing, Quality Control and Leakage-Free Protocol

- **Profiling and integrity**: formats, pseudonymized identifiers, duplicates, missing or corrupted images, metadata consistency.
- **Spatial preprocessing**: breast region by Otsu thresholding, background reduction, resizing to 224 × 224 and normalization.
- **Neutralization of inscriptions**: text on ultrasounds located by EasyOCR, masked then reconstructed by Navier–Stokes-type inpainting.
- **Partitioning**: five-fold **GroupKFold** cross-validation by patient; the Clinical RAG additionally excludes the current record and the documents of its fold.

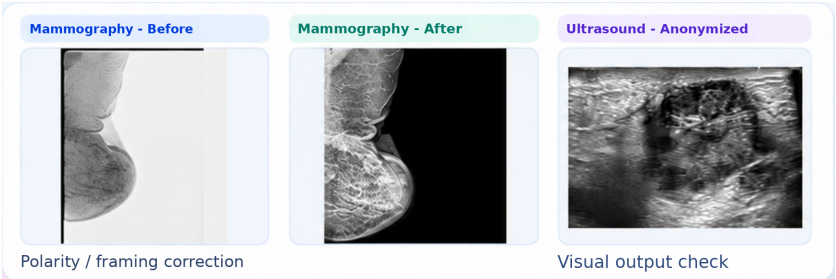


Figure 2.2: Before/after examples of preparation and pseudonymization.

Methodological precaution. A correct image-level split is not enough when a patient has several examinations. Patient-level separation and RAG document isolation are two distinct measures against information leakage, to be audited at each reproduction.

2.3 ResNet-18 Models and Multimodal Fusion

Each modality has a ResNet-18 encoder [3] with a single-channel 224 × 224 input, which accounts for the difference in texture and acquisition. Four fusion configurations are described: **late fusion** (score averaging), **weighted logit fusion**, **uncertainty-guided fusion** (weighting inversely related to entropy) and the **multimodal ensemble**, which combines the previous ones with test-time augmentation (TTA). Shannon entropy $H(p) = -\Sigma\ p_c\ log\ p_c$ serves as an uncertainty indicator; it does not turn scores into clinically calibrated probabilities. The comparison covers 323 patients (GroupKFold, “malignant versus others”) with no undocumented hyperparameter search.

2.4 Explainability with Grad-CAM

Grad-CAM [4] relies on the gradients of a target class with respect to feature maps: $\alpha_k^c = (1/Z)\ \Sigma_{i,j} \partial y^c / \partial A_{ij}^k$ and $L^c = ReLU(\Sigma_k\ \alpha_k^c\ A^k)$. The audit covers 387 maps associated with 64 records: the mean energy inside the breast mask reaches **87.4%** (12.6% outside). This supports spatially plausible attention, without demonstrating a systematic correspondence with lesions.

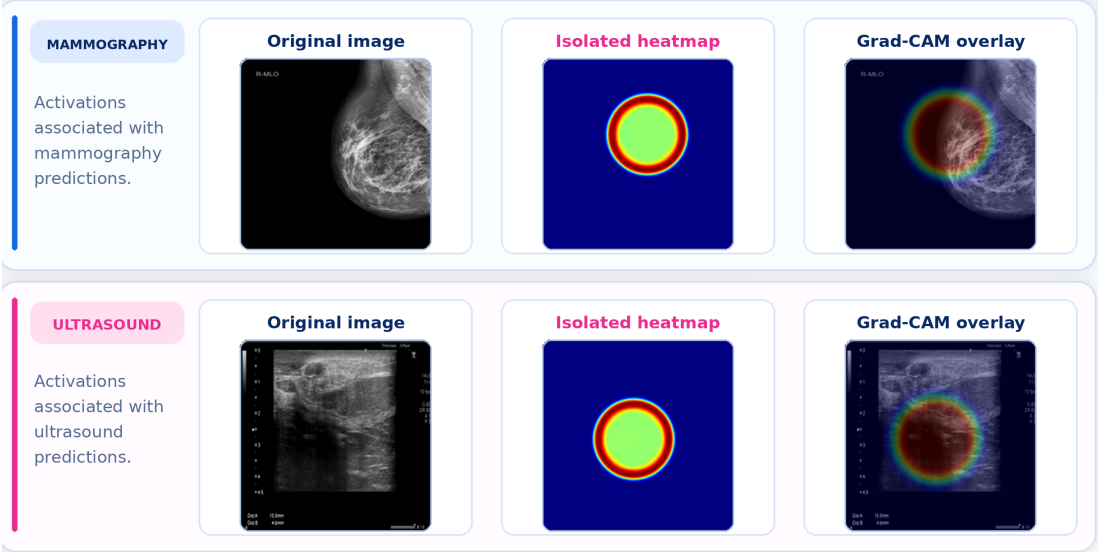


Figure 2.3: Original views, activation maps and Grad-CAM overlays for both modalities.

Link between vision and document agents. The vision agent transmits logits, probabilities, uncertainties and visualizations; the document and generative agents must not alter these values. A good ROC-AUC and a plausible map replace neither a histological reference nor a radiologist’s opinion: hence the independent verification agent before any output.

CHAPTER 3

Clinical RAG and Prompt Engineering

3.1 Knowledge Base and Semantic Search

The Clinical RAG retrieves passages describing relevant rules and observations so that the generative engine justifies its wording from an explicit context; it documents the synthesis without certifying its medical accuracy. Documents are extracted, cleaned, segmented, encoded by **BAAI/bge-small-en-v1.5** (384 dimensions) and indexed in **FAISS IndexFlatIP** (inner product of unit vectors, equivalent to cosine similarity [5, 6]). Two indexes are maintained:

- **Guidelines index**: nine prototype rules inspired by BI-RADS principles (non-exhaustive, unofficial);
- **Case index**: 320 anonymized reports (one of 321 excluded), i.e. 995 passages.

Segment sizes in tokens and overlap are not documented and are therefore not presented as validated. Search returns up to **four passages** identified by [G1] or [E1], whose validity is checked downstream. Two filters limit leakage: **text_blind** (removal of mentions of BI-RADS category, confirmed cancer or treatment performed) and **exclude_pseudo_id** (removal of the evaluated patient and their fold). Their effectiveness must be confirmed by checks at each fold.

3.2 Choice and Integration of Qwen2.5-3B-Instruct

The selected LLM [7] is motivated by its size, its local execution and its ability to follow an imposed format (about 3.1 GB occupied on a Tesla T4 GPU in a project notebook, a value that depends on the configuration). It was **not fine-tuned**: adaptation relies on prompt engineering and documentary context, whereas supervised learning concerns the vision branches. Zero-temperature decoding limits variability without guaranteeing absolute determinism. No quantitative benchmark between competing LLMs is available.

3.3 Design of Structured Medical Prompts

Part	Information included	Targeted constraint
SYSTEM	Synthesis role, neutrality, ban on inventing observations	Limit unsupported claims, enforce JSON
CONTEXT	Scores, uncertainty, Grad-CAM summary, authorized context, RAG excerpts	Separate computed facts from documentary sources
TASK	Reconciliation of modalities, interpretation and proposed follow-up	Synthesis consistent with the context
OUTPUT	Required fields and expected types	Automatic validation

The output schema requires six fields: `predicted_birads`, `confidence`, `interpretation`, `recommendations`, `limitations`, `human_review_required`. The `confidence` field is a prototype indicator, not a validated clinical probability. A set of **190 prompts** is documented (64 image only, 62 text only, 64 image + text); these numbers describe how the trials were constructed and are not enough to conclude that one condition is superior.

RAG evaluation. No quantified table of precision, Recall@4 or chunking comparison is provided in the source report; no value is invented. Nine rules and 995 passages do not constitute a validation of retrieval quality. The history of prompt versions must be presented separately, based on laboratory traces.

CHAPTER 4

Automation, Validation and Results

4.1 FastAPI Services and n8n Workflows

Processing is exposed through FastAPI services; n8n links the steps, applies routing conditions and keeps a trace of executions. Two workflows appear in the sources: **BCMID-Medical-Analysis** (main chain of eight nodes, from the webhook to the response with results and report) and **Triage** (nine nodes: reception, identifier extraction, call to the report service, conditional branching, output). Triage relies on a logical field `requires_immediate_escalation`; the urgent branch corresponds to the heuristic BI-RADS 4A to 5 classes. These actions do not demonstrate a real clinical integration with a PACS/HIS.

4.2 Control Loop and Automatic Correction

The Reasoning ↔ Quality & Safety coupling is central: the LLM's proposal is subjected to deterministic validations that are independent of its claims. In case of inconsistency, Agent 6 refuses validation and produces a targeted corrective instruction; **two iterations** are allowed, after which the `FLAGGED_FOR_MANUAL_CLINICIAN_REVIEW` state is triggered, without forcing a report to be presented as compliant.

Check	Control rule
Numerical consistency	Tolerated gap between cited probability and model probability: $\|\Delta p\| \leq 0.006$
Uncertainty	Beyond the prototype threshold $H \geq 0.50$, require cautious wording
Clinical entities	Reject observations not attested by vision or validated texts
Citations	Each reference must match a passage retrieved in the request context
Format / supervision	Check explicit limitations; enforce human_review_required = true

These rules frame identified behaviors but do not prove the absence of any error: a false statement can be well structured, and cited content can be out of context.

4.3 Safety, Data Protection and Regulatory Limits

A table converts the continuous malignancy score into a category and a workflow priority. It is **strictly heuristic and not clinically validated**; it must neither be confused with the ACR BI-RADS diagnostic definition, nor guide real patient management without a radiologist.

Score p_{mal}	Indicative category	Internal priority
$p < 0.02$	BI-RADS 1	Routine
$0.02 \leq p < 0.10$	BI-RADS 2	Routine
$0.10 \leq p < 0.35$	BI-RADS 3	Surveillance
$0.35 \leq p < 0.65$	BI-RADS 4A	Escalation planned in the workflow
$0.65 \leq p < 0.85$	BI-RADS 4B	Escalation planned in the workflow
$0.85 \leq p < 0.95$	BI-RADS 4C	Escalation planned in the workflow
$p \geq 0.95$	BI-RADS 5	Escalation planned in the workflow

The pipeline provides for pseudonymization of identifiers using **SHA-256 with a local salt** and masking of ultrasound inscriptions; local operation of FAISS and the LLM reduces reliance on the cloud. Pseudonymization is nevertheless not definitive anonymization: key management, access, logs and retention periods remain indispensable. The GDPR, HIPAA and, in the event of commercialization in Europe, Regulation MDR 2017/745 [8] are frameworks to be examined; the project provides neither a full compliance audit nor certification.

4.4 Multimodal Classification Performance

The measurements do not concern the same objects: classification (323 patients), XAI audit (387 maps) and resilience tests (software validation cases), with distinct samples and criteria.

Configuration	ROC-AUC	Sensitivity
Mammography alone	0.7917	76.2%
Ultrasound alone	0.6743	68.4%
Late fusion	0.8150	81.5%
Weighted logit fusion	0.8160	81.8%
Uncertainty fusion	0.8120	80.9%
Multimodal ensemble with TTA	**0.8198**	**83.1%**

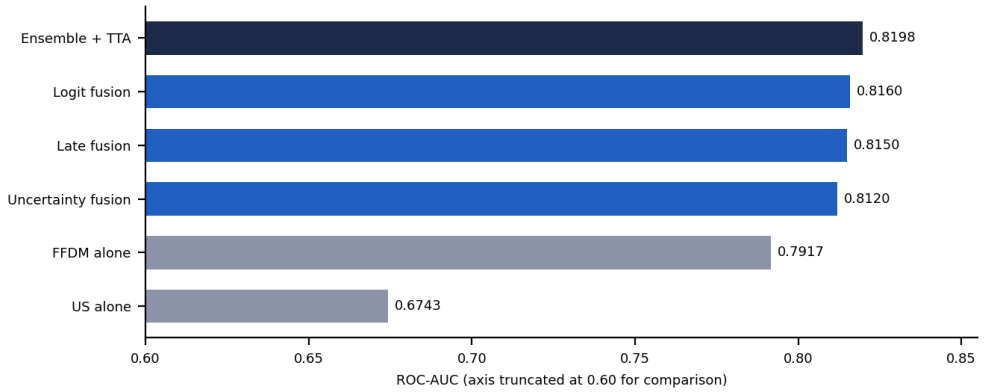


Figure 4.1: Comparison of declared ROC-AUC values (axis truncated at 0.60).

The ensemble exceeds mammography alone by 0.0281 ROC-AUC points and ultrasound alone by 0.1455; sensitivity rises from 76.2% to 83.1%. Differences between fusion methods are limited and **no significance test or confidence interval** is documented.

4.5 Audit of the Agentic Chain and Case Study

Item	Result	Scope of the measurement
Prompts built	190	64 image only; 62 text only; 64 image + text
Adversarial tests	10/10	Injection and contradictory instructions, limited sample
Reports compliant at 1st check	17/20	3 reports to review after the first pass (85%)
n8n executions	23	5 failures (21.7%) reported in the tracking
Average duration displayed	5.54 s	n8n log indicator; scope to be documented
Successful full execution	12.984 s	Observed example, not a statistical measurement
pytest tests	61/61	Passed in 1.84 s in the environment shown

Case study. The pseudonymized record `Paacb1ec57b` (mammography and targeted ultrasound) yields a category derived from the prototype's BI-RADS 3 logic, p_{mal} = 0.12, displayed confidence of 78%, entropy H = 0.38 and 89.2% of Grad-CAM energy within the mask. This is an example of prototype output, to be validated by a professional.

4.6 Critical Discussion and Threats to Validity

Structural limitations. • **Cohort**: 332 patients, class imbalance, 323 for the multimodal comparison. • **Ground truth**: labels derived from BI-RADS without histological verification. • **Generalization**: no external multicenter or prospective validation. • **Comparisons**: no confidence interval, no statistical test between fusions, no benchmark between LLMs. • **XAI**: 12.6% of Grad-CAM energy outside the breast region. • **Integration**: 5 n8n failures out of 23; 3 reports out of 20 to be revised from the first pass. • **Medical decision**: score–BI-RADS and uncertainty thresholds not clinically validated.

General Conclusion

The project assembles complementary contributions: integration of two imaging modalities, comparative evaluation of fusion strategies, XAI elements, a local vector document base and the framing of an LLM's outputs by independent rules. The strength of the system lies less in any single component than in the **separation of responsibilities** between prediction, document retrieval, writing and validation. Fusion achieves a ROC-AUC higher than that of each modality alone (0.8198), Grad-CAM provides spatial information, the RAG gives access to documentary excerpts, and the reasoning and safety agents structure and then check the report.

Perspectives. External validation with independent clinical references; audit of n8n workflow stability; quantitative evaluation of the RAG (Recall@4, citation accuracy) and of prompt iterations; score calibration; examination of critical errors and specialized verification of triage rules before any extension to real medical environments. This contribution remains that of an academic prototype, in which every output is traceable, every uncertainty flagged and every interpretation subject to human expertise.

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